

RDFS Knowledge Graphs Through the Lens of Logic: Semantics and Query Answering

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Abstract. We propose a semantic characterization for a simple and yet sufficiently expressive kind of Knowledge Graphs based on RDFS, which we call RKGs. In doing so, we address some of the issues deriving from the standard RDFS semantics and then proceed by defining a semantics for RKGs based on formal logic which is also able to properly capture the metamodeling capabilities of RDFS. We prove that RKGs, in general, do not exhibit a canonical model, and thus query answering cannot rely on the well-known techniques based on the use of it. Then, by singling out that the lack of a canonical model for RKGs is related to the presence of a form of indefiniteness in RKG elements, we define the class of *definite RKGs*, and prove that definiteness is both a sufficient and necessary condition for an RKG to admit a canonical model. Also, based on such result, we propose an algorithm for query answering for both definite and general RKGs and we characterize the computational complexity in both cases.

1 Introduction

Recent developments in Artificial Intelligence have contributed to increasing the interest in Knowledge Graphs (KGs), both in research and industry. Their relevance stems from their various applications within AI systems. Two notable examples are Data Preparation and Neuro-Symbolic AI, the first concerning the set of processes of collecting, cleaning, and transforming data to make it ready for use in AI systems based on Machine Learning or for analysis, the latter regarding those approaches trying to combine symbolic and sub-symbolic techniques in AI systems with the focus of enhancing both their performance and reliability. In both cases, KGs represent a valuable tool, as by encapsulating both intensional and extensional knowledge, they enable to collect, manage and integrate large amounts of heterogeneous data, while also providing the main reasoning capabilities, such as query answering, consistency checking, or the derivation of new knowledge, by means of either deductive or inductive techniques (the latter being based on the graphs' topology).

Background on Knowledge Graphs. Nowadays, several different frameworks exist for creating and managing KGs [16]. In some contexts, KGs coincide with plain graph databases [1] equipped with query languages typically based on pattern-matching mechanisms. This kind of approach, typical of, e.g., property graphs databases [18] such as Neo4j and Sparksee, limits their usefulness by excluding the

semantic aspects that allow for carrying out reasoning tasks over the data. A different approach is the one where KGs are represented as Description Logic (DL) knowledge bases (or ontologies) [17, 5, 6]. Such formalisms provide KGs with a proper logic-based semantics, but they still present some limitations, in that they fall short in properly capturing metamodeling features. Indeed, most works on DLs are typically based on treating metamodeling features by means of punning, according to which two occurrences of the same domain element in different positions are treated as if they refer to different elements. A notable exception is the proposal of a proper metamodeling semantics for OWL 2 QL ontologies reported in [19]; however, to the best of our knowledge there are still no systems implementing such framework.

Another possible approach, which is also one of the most well-known and widely used, is that of KGs based on RDFS, a framework derived from the Semantic Web domain. An RDFS graph consists of a set of triples of the form $\langle s \ p \ o \rangle$, each representing an edge of type p connecting the two nodes s and o . RDFS graphs rely on a vocabulary of symbols endowed with semantics, enabling it to capture the meaning of some fundamental relationships between entities, such as membership assertions, or subset relations between concepts.

Although RDFS graphs serve as a useful tool for KG management and are capable of capturing forms of metamodeling (i.e., they allow for representing domains where elements can simultaneously play different roles, such as that of a class and of an individual, or that of a property and of a class), and although open source tools for managing RDFS knowledge graphs are currently available, such as Ontotext GraphDB¹ and Ontop [6], several issues with the semantics of RDFS graphs have been singled out. A first series of papers [22, 20] have been devoted to correct the set of so-called *entailment rules*, which were originally provided as a mean to capture the “official” RDFS semantics², also called *intensional* semantics, and were shown to be incomplete. More importantly, several works have pointed out severe drawbacks of the intensional semantics (see, e.g., [11]). In particular, such semantics fails to define sets in terms of their extensions, thus not being able to fully capture the set-theoretic notions underlying basic constructs such as the subset relation (e.g., the one corresponding to the `subClassOf` construct). These limitations affect the significance of the represented knowledge and the compatibility with most widely used, logic-based knowledge representation languages, such

¹ <https://www.ontotext.com/products/graphdb/>

² <http://www.w3.org/TR/rdf-schema/>

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as OWL³. Also, all the works adopting the intensional semantics address query answering over RDFS graphs by resorting to the RDFS entailment regime [2, 12, 7, 21]. According to this regime, the existential variables in queries are not treated as in first-order classical logic. Indeed, a correct binding for such variables associates to each variable the same domain object in every model, while the classical logic semantics look for the existence of a binding in every model, possibly accepting different bindings in different models.

To partially address the limitations of the intensional semantics, RDFS was provided with a “non-normative” semantics, called *extensional* [15], which in fact corresponds to the classical logic semantics, together with a set of additional entailment rules capturing such semantics. In [11], the authors propose a proof-theoretic technique for RDFS entailment under the extensional semantics based on an extended set of entailment rules. Note that neither [15], nor [11] study the problem of query answering under the extensional semantics. In [10], the authors use a model-theoretic approach and study the logical foundations of both the intensional and extensional semantics of RDFS. To this aim, they adopt a technique based on embedding RDFS entailment into F-Logic, to reason over RDFS graphs by means of Datalog engines. To the best of our knowledge, [10] is the only work considering query answering under the classical logic, even though not for the extensional semantics of RDFS. Taking into account all the related work just mentioned, we can conclude that the problem of developing algorithms and complexity analyses for query answering over RDFS knowledge graphs when both the graph and the query are interpreted according to classical logic is open.

Contributions. With the goal of defining a well-founded, logic-based semantic characterization for KGs including both metamodeling and (meta)querying, in this paper we consider KGs expressed in RDFS (extended with the possibility of using blank nodes in any position), that we call RDFS KGs (RKGs), and provide the following main contributions.

- Inspired by [19], we propose a comprehensive semantic framework, that we simply call *RKG Metamodeling Semantics*, for interpreting and querying RKGs based on classical logic. Queries are expressed using the SPARQL syntax, but contrarily to the official semantics of SPARQL, in our framework, we interpret queries by resorting to the classical logic interpretation of their construct, in particular conjunction and existential quantification.
- We study the logical properties of RKGs in our framework and show that, differently from what one might expect, in general RKGs do not exhibit canonical models. This implies that query answering cannot rely on the well-known techniques based on their use. Then, by singling out that the lack of a canonical model for RKGs is related to the presence of a form of indefiniteness in the graph elements, we define the class of definite RKGs, and prove that definiteness is both a sufficient and necessary condition for an RKG to admit a canonical model.
- We present an algorithm for query answering under the proposed semantic framework both in definite and in general RKGs and, finally, characterize the computational complexity of query answering in both cases. Surprisingly, it turns out that the complexity of query answering in our framework is Π_2^P for general RKGs.

As far as we are aware, our work is the first studying query answering in RDFS-based knowledge graphs under a metamodeling semantics that interprets both the KG and the queries in terms of classical logic.

Plan of the paper. The paper is organized as follows. In Section 2 we present the syntax of RKGs and the basic notions related to the query language. In Section 3 we introduce the RKG Metamodeling Semantics framework and define query answering under the proposed semantics and in Section 4 we discuss the logical properties of RKGs in the new framework. Then, in Section 5, we illustrate the chase procedure. Finally, in Sections 5 and Section 6, we study algorithms and complexity for query answering in definite and general RKGs.

2 Preliminaries

In this section we recall the syntax of RDFS graphs and SPARQL queries.

An RDFS graph is a directed edge-labeled graph, i.e., a set of vertices connected by directed edges with an associated label, such that vertices and edges are called *resources* and are labeled by resource names, that can be of one of three types, namely, IRIs, literals, or blank nodes. IRIs denote objects in the so-called universe of discourse, literals denote values belonging to some datatype defining the range of possible values, and blank nodes are used as placeholders to denote the existence of some resource whose name is not known. Intuitively, a directed edge labeled with p , from a vertex labeled with a to a vertex labeled with b expresses that the object denoted by a is related to the object denoted by b through the binary relation denoted by p . Further distinguishing features of RDFS graphs concern the syntax and the semantics, respectively. As for the syntax, coherently with RDF, which RDFS extends, literals and blank node names can label vertices only, and literals only vertices which are not source of any edge. It follows that in RDFS graphs edges can be labeled by IRIs only. As for the semantics, IRIs belonging to a “reserved” vocabulary, the so-called *RDFS vocabulary*, come with a “special” meaning. Thus, for example, `rdfs:Class` is used to denote the set of objects which are themselves sets of objects, `rdf:type` the relationship of membership between an object and a certain class, and `rdfs:subClassOf` the subset relationship between classes.

In this work, we consider a specific type of graphs derived from RDFS graphs that we call *RKGs*, comprising a subset of the RDFS vocabulary and not containing any literal. This is due to the fact that we aim at investigating the semantics and computational properties of the core of RDFS, which we argue make it a basic yet effective tool for modeling a domain of interest. Specifically, we assume that the RDFS vocabulary $\mathbb{R}$ is such that it consists only of the following IRIs⁴: `Resource`, `Class`, `Property`, `type`, `domain`, `range`, `subClassOf`, and `subPropertyOf`. Coherently with the literature on the domain, in order to represent RKGs, we resort to sets of *RKG triples* of *elements* denoted by symbols from an alphabet $\mathcal{A} = \mathbb{I} \cup \mathbb{B}$, with $\mathbb{I}$ being a set of IRIs including $\mathbb{R}$, and $\mathbb{B}$ being a set of symbols, disjoint from $\mathbb{I}$, denoting blank nodes (we assume that symbols in $\mathbb{B}$ start with “_”). Thus, an *RKG triple*, or simply *triple*, has the form

$$\langle a \ b \ c \rangle$$

with elements $a, b, c \in \mathcal{A}$, a being the *subject* of the triple, b being the *predicate*, and c being the *object*. Notice that our syntax generalizes the standard RDF syntax, in that we allow blank nodes to appear in property position. Finally, given an RKG G , we denote by $\mathbb{I}^G$ and $\mathbb{B}^G$ the subsets of $\mathbb{I}$ and $\mathbb{B}$ occurring in G .

³ <https://www.w3.org/TR/owl2-direct-semantics/>

⁴ For the sake of clarity, we omit IRIs namespace prefixes.

In our discussion, we rely on the triple-based representation just described, and we use the terms *triple* and *axiom* interchangeably, since, from a logical point of view, an RKG can be considered as a Knowledge Base with the triples expressing its axioms. Clearly, RKG triples can be graphically represented as directed edges connecting a source node, corresponding to the subject of a triple, to a target node corresponding to the object of a triple. Such edges are labeled as the property of the corresponding triple. The following example illustrates an RKG expressed by means of a set of RDFS triples.

Example 1. The following set of triples represents the set of axioms of a valid RKG. The graph models a scenario where someone, whose name is not known, teaches to Alice, those to whom it is taught are students, those who teach are professors, professors are persons, and professor is a faculty role. Note that the relationship between the class Professor and the class FacultyRole is of type *instance-of*, i.e., the FacultyRole class is indeed a metaclass, since it has other classes among its instances.

```
{ (:b teachesTo Alice),
  (teachesTo range Student),
  (teachesTo domain Professor),
  (Professor subClassOf Person),
  (Professor type FacultyRole) }
```

Fig. 1 shows the graph representation of the above RKG. Notice that the graph is constituted by five edges, one for each of the axioms of the example. The dashed line connecting the label teachesTo to the node teachesTo is added for the sake of clarity to specify that the node refers to the label of the edge. This derives from the metamodelling capabilities of RDFS which allow to predicate on properties. In particular, if we need to predicate over a certain property, i.e., having it to appear either in subject or object position in some triple, then we need to introduce a node for it, as in the depicted example.

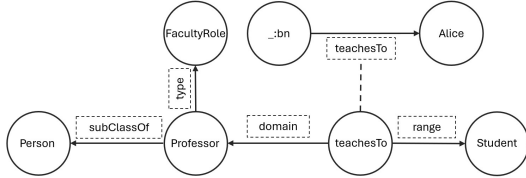


Figure 1: Graphical representation of the RKG from Example 1.

We now briefly recall the basic notions of the SPARQL query language, as we adopt its syntax when dealing with query answering. SPARQL⁵ is a declarative language which allows retrieving data from RDF graphs by specifying patterns that the queried data must satisfy. In its simplest form, a SPARQL query consists of a conjunction (denoted by symbol ".") of triples, each one representing a template, possibly including variables (variable names in SPARQL begin with ? or with \$), that should match the data present in the RDFS graph. Such simple patterns are called Basic Graph Patterns (BGP). When answering a SPARQL query over an RDFS graph, the query patterns are evaluated against the graph obtained by applying a suitable entailment regime, i.e., a set of inference rules used to complete a graph with the knowledge that can be derived from known facts, to retrieve matching subgraphs. The evaluation of the patterns is based on the notion of variable binding, that is the operation of finding a matching between variables and data satisfying the patterns.

Example 2. Consider the following SPARQL query evaluated against the RKG of Example 1:

ASK WHERE { ?x type Person. }

The above boolean query returns *true* if and only if there exists an assignment for the variable x such that the triple in the WHERE clause is satisfied by the underlying graph.

Based on the RDFS entailment regime, the query returns *true* as an answer, as the binding $x \leftarrow _ : b$ satisfies the query. This is due to the fact that, according to the RDFS entailment regime, the element represented by the blank node denoted as $_ : b$ is an instance of Professor, and since Professor is a subclass of Person, then the element denoted by $_ : b$ is also an instance of the class Person, which satisfies the query pattern.

Query Entailment In this paper, we study the computational complexity of the problem of *query entailment*, i.e., deciding whether an RKG G logically implies a conjunctive query Q . In particular, we study the *combined complexity* of the problem, i.e., where both G and Q are provided as input.

3 The RKG Metamodeling Semantics

In this section, we define a comprehensive framework for interpreting and Querying RKGs, which we call RKG Metamodeling Semantics [19].

An *interpretation* $\mathcal{I}$ for an RKG G is a pair $\langle \mathcal{W}, \cdot^{\mathcal{I}} \rangle$, where $\mathcal{W}$ is the *interpretation structure* of $\mathcal{I}$ and $\cdot^{\mathcal{I}}$ is the *interpretation function* of $\mathcal{I}$. In particular,

- $\mathcal{W}$ is a triple $\langle \Delta, \cdot^C, \cdot^P \rangle$ such that:
 - Δ is a non-empty set of objects, called the domain of $\mathcal{I}$;
 - $\cdot^C : \Delta \rightarrow \mathcal{P}(\Delta)$ is a partial function;
 - $\cdot^P : \Delta \rightarrow \mathcal{P}(\Delta \times \Delta)$ is a partial function.

Intuitively, function $\cdot^C$ (resp., $\cdot^P$) is defined for the objects of the domain that are seen as a class (resp., property) in $\mathcal{W}$, and determines the class extension (resp., property extension) for each of them. Indeed, we say that $o \in \Delta$ *plays the role of class* (resp. *property*) in $\mathcal{I}$ if o^C (resp. o^P) is defined.

- $\cdot^{\mathcal{I}}$ maps every symbol in $\mathbb{I}^G$ into an object in Δ and therefore sanctions how to interpret the various symbols in terms of the structure $\mathcal{W}$. In doing so, $\cdot^{\mathcal{I}}$ ensures that all predefined symbols are interpreted correctly according to their intended meaning, namely:

- $(\text{Resource}^{\mathcal{I}})^C = \Delta$;
- $(\text{Class}^{\mathcal{I}})^C = \{ o \in \Delta \mid \cdot^C \text{ is defined for } o \}$,
- $(\text{Property}^{\mathcal{I}})^C = \{ o \in \Delta \mid \cdot^P \text{ is defined for } o \}$,
- $(\text{type}^{\mathcal{I}})^P = \{ (o_1, o_2) \mid o_1, o_2 \in \Delta \text{ and } o_1 \in o_2^C \}$,
- $(\text{subClassOf}^{\mathcal{I}})^P = \{ (o_1, o_2) \mid o_1, o_2 \in \Delta \text{ and } o_1^C \subseteq o_2^C \}$,
- $(\text{subPropertyOf}^{\mathcal{I}})^P = \{ (o_1, o_2) \mid o_1, o_2 \in \Delta \text{ and } o_1^P \subseteq o_2^P \}$,
- $(\text{range}^{\mathcal{I}})^P = \{ (o_1, o_2) \mid \{ o_3 \mid \exists o_4 (o_4, o_3) \in o_1^P \} \subseteq o_2^C \}$,
- $(\text{domain}^{\mathcal{I}})^P = \{ (o_1, o_2) \mid \{ o_3 \mid \exists o_4 (o_3, o_4) \in o_1^P \} \subseteq o_2^C \}$.

For example, the first item specifies that the symbol Resource is interpreted by $\mathcal{I}$ as the object for which $\cdot^C$ is defined (and therefore is a class) and its class extension coincides with the whole domain of $\mathcal{I}$. Analogously, the second item specifies that the symbol Class is interpreted by $\mathcal{I}$ as the object for which $\cdot^C$ is defined and its class extension coincides with the objects of the domain of $\mathcal{I}$ that play the role of a class.

⁵ <https://www.w3.org/TR/2013/REC-sparql11-query-20130321/>

In order to deal with the presence of blank nodes in RKGs, we also introduce an *assignment function for an RKG G coherent with an interpretation $\mathcal{I}$* , denoted $\nu_{\mathcal{I}}$, which is such that:

$$\nu_{\mathcal{I}}(x) = \begin{cases} x^{\mathcal{I}} & \text{if } x \in \mathbb{I}^G \\ o \text{ (for some } o \in \Delta) & \text{if } x \in \mathbb{B}^G \end{cases}$$

It is worth noting that according to the semantics defined above, it is possible that an element $e \in (\mathbb{I} \cup \mathbb{B})$ of the KG is mapped by $\mathcal{I}$ to an object of Δ that, according to $\mathcal{W}$, simultaneously plays the role of a class and that of a property. In this sense, the element e is modeled in G as an individual that can also be a class, a property, or both. This captures the notion of metamodeling, as present, for instance, in RDFS.

Now, the semantics of an RKG G is defined in terms of satisfaction of a triple with respect to both an interpretation $\mathcal{I}$ for G and an assignment $\nu_{\mathcal{I}}$ coherent with $\mathcal{I}$. We say that $\mathcal{I}$ *satisfies a triple $\langle a, b, c \rangle$ under $\nu_{\mathcal{I}}$* , denoted as $(\mathcal{I}, \nu_{\mathcal{I}}) \models \langle a, b, c \rangle$, if $(\nu_{\mathcal{I}}(a), \nu_{\mathcal{I}}(c)) \in \nu_{\mathcal{I}}(b)^P$. Also, $(\mathcal{I}, \nu_{\mathcal{I}}) \models G$ means $(\mathcal{I}, \nu_{\mathcal{I}}) \models \langle a, b, c \rangle$ for every triple $\langle a, b, c \rangle \in G$ and $\mathcal{I} \models G$ means that $\mathcal{I}$ is a *model* of G , i.e., if there exists an assignment $\nu_{\mathcal{I}}$ coherent with $\mathcal{I}$ such that $(\mathcal{I}, \nu_{\mathcal{I}}) \models G$. We denote by $Mod(G)$ the set of models of G , and we say that G entails a triple α , written $G \models \alpha$, if $\mathcal{I} \models \alpha$ for every $\mathcal{I} \in Mod(G)$. Since G does not contain any form of negation, it is immediate to verify that $Mod(G) \neq \emptyset$ for every RKG G .

We now turn our attention to querying RKGs, for which we resort to the SPARQL syntax introduced in the previous section. In particular, we consider the subset of SPARQL queries that consists of BGPs, where a BGP, as we said, is a conjunction of triples, possibly containing variables in any position. Note that since blank nodes act as variables, without loss of generality, we assume that triples in BGPs have the same form as triples in RKGs: in other words, we express variables in the queries by means of blank nodes. Also, for the sake of brevity, we consider only boolean SPARQL queries and we do not consider unions of BGPs. We point out, however, that all the results presented in the paper can be easily extended to the case of non-boolean unions of BGPs.

Formally, as far as this paper is concerned, a *query Q* is an expression of the form $ASK \text{ WHERE } \{B\}$, where B , called the *body* of Q , is a BGP, i.e., a set of triples. Also, we denote by $\mathbb{I}^Q$ and $\mathbb{B}^Q$, the subsets of the symbols of $\mathbb{I}$ and $\mathbb{B}$ occurring in B .

We next characterize the semantics of queries in the RKG Metamodeling Semantics framework. In doing so, we follow our goal of remaining faithful to logic in the interpretation of disjunction (UNION), conjunction and existential quantification. First, we define when a query is true in a given interpretation of an RKG G , and then we define when a query is entailed by G .

Given a query Q expressed over an RKG G , and an interpretation $\mathcal{I} = \langle \mathcal{W}, \cdot^{\mathcal{I}} \rangle$ of G , the notion of Q being true in $\mathcal{I}$ is the usual one in first-order logic, based on interpreting blank nodes in the query as existentially quantified variables: Q is true in $\mathcal{I}$, written $\mathcal{I} \models Q$, if there exists an assignment that associates to each blank node in B an object in Δ such that the resulting triples in Q are all satisfied by $\mathcal{I}$. It is easy to see that this can be formalized by the notion of query homomorphism as follows.

Definition 1. Let G be an RKG, $\mathcal{I} = \langle \mathcal{W}, \cdot^{\mathcal{I}} \rangle$ an interpretation for G with domain Δ , and Q a BGP. A query homomorphism from Q to $\mathcal{I}$ is a total function Ψ from $(\mathbb{I}^Q \cup \mathbb{B}^Q)$ to Δ , such that:

- for every $s \in \mathbb{I}^Q$, $\Psi(s) = s^{\mathcal{I}}$,

- for every $\langle s_1, s_2, s_3 \rangle \in Q$, $\langle \Psi(s_1), \Psi(s_2) \rangle \in \Psi(s_3)^P$.

It is immediate to see that $\mathcal{I} \models Q$ if and only if there exists a query homomorphism from Q to $\mathcal{I}$. Based on this property, we can finally establish the semantics of a query posed to RKG. Since we focus on boolean queries, query answering reduces to query entailment, that checks whether the query is true in every model of the RKG (certain answer semantics).

Definition 2. An RKG G entails a query Q , denoted $G \models Q$, if for every $\mathcal{I} \in Mod(G)$, $\mathcal{I} \models Q$ holds, i.e., there exists a query homomorphism from Q to $\mathcal{I}$.

The above definition makes it clear that, in order to conclude that $G \models Q$ we follow classical logic, and check if for every model $\mathcal{I}$ of G there is a binding for the existential variables of Q that makes the query true in $\mathcal{I}$. Note the difference with the SPARQL entailment regime, where, instead, one checks if there exists a binding for the existential variables of Q that makes the query true in every model of G . This switch of quantifiers marks the difference between the SPARQL entailment regime and our notion, based on classical logic.

4 Logical properties of RKGs

In this section we investigate the logical properties of RKGs derived by our formalization using the Metamodeling Semantics. To this aim, we introduce the notion of “canonical model” in our context and show that differently from other apparently similar formalisms, RKGs do not admit any such model. Then, guided by this result, we introduce a restricted class of RKGs, which we call “definite”.

In order to pursue our logical characterization of RKGs, we next define the notion of RKG homomorphism.

Definition 3. Let $\mathcal{I}_1 = \langle \mathcal{W}_1, \cdot^{\mathcal{I}_1} \rangle$ and $\mathcal{I}_2 = \langle \mathcal{W}_2, \cdot^{\mathcal{I}_2} \rangle$ be two interpretations for an RKG G with $\mathcal{W}_1 = \langle \Delta_1, \cdot^{C_1}, \cdot^{P_1} \rangle$ and $\mathcal{W}_2 = \langle \Delta_2, \cdot^{C_2}, \cdot^{P_2} \rangle$. An RKG homomorphism from $\mathcal{I}_1$ to $\mathcal{I}_2$ is a total function Ψ from Δ_1 to Δ_2 such that:

- for every $e \in \mathbb{I}$ and $o \in \Delta_1$, if $e^{\mathcal{I}_1} = o$, then $e^{\mathcal{I}_2} = \Psi(o)$;
- for every $o_1, o_2, o_3 \in \Delta_1$, if $\langle o_2, o_3 \rangle \in o_1^{P_1}$, then $\langle \Psi(o_2), \Psi(o_3) \rangle \in \Psi(o_1)^{P_2}$.

Intuitively, an RKG homomorphism from an interpretation $\mathcal{I}_1$ to an interpretation $\mathcal{I}_2$ is a transformation from the former to the latter, such that the semantic properties of $\mathcal{I}_1$ are preserved in $\mathcal{I}_2$. With the above notion in place, we are ready to define canonical models for RKGs.

Definition 4. Let G be an RKG and let $\mathcal{I}$ be a model of G . We say that $\mathcal{I}$ is a canonical model of G (or, simply, canonical) if, for every model $\mathcal{J}$ of G , there exists an RKG homomorphism from $\mathcal{I}$ to $\mathcal{J}$.

It follows from the definition that a canonical model $\mathcal{I}$ of an RKG G is such that every triple that is true in $\mathcal{I}$, is also true in every model of G . In other words, $\mathcal{I}$ is “semantically representative” of every model of G .

We next present one of the main results of this paper: RKGs do not generally admit a canonical model.

Proposition 1. There exists an RKG G such that no interpretation of G is a canonical model.

Proof. Let G be a RKG such that

- $a, b \in (\mathbb{I} \cup \mathbb{B})$,

- $G \not\models \langle x \text{ type } a \rangle$, for every $x \in (\mathbb{I} \cup \mathbb{B})$,
- $G \not\models \langle a \text{ subClassOf } b \rangle$,
- $G \models \langle a \text{ type Class} \rangle, G \models \langle b \text{ type Class} \rangle$.

Note that one such G exists, for example, $G_1 = \{\langle a \text{ type Class} \rangle, \langle b \text{ type Class} \rangle\}$. Now, let $\mathcal{I}_0 = \langle \mathcal{W}_0, \cdot^{\mathcal{I}_0} \rangle$ be any model of G , with $\mathcal{W}_0 = \langle \Delta_0, \cdot^{C_0}, \cdot^{P_0} \rangle$. We show that $\mathcal{I}_0$ is not a canonical model. Actually, there are two cases to consider for $\mathcal{I}_0$: either $(a^{\mathcal{I}_0})^{C_0} = \emptyset$, or $(a^{\mathcal{I}_0})^{C_0} \neq \emptyset$.

1. The case of $(a^{\mathcal{I}_0})^{C_0} = \emptyset$.

Note that this case can indeed occur, because $G \not\models \langle x \text{ type } a \rangle$, for every $x \in (\mathbb{I} \cup \mathbb{B})$ and it obviously implies $\mathcal{I}_0 \models \langle a \text{ subClassOf } b \rangle$. Now, consider a model $\mathcal{I}_1 = \langle \mathcal{W}_1, \cdot^{\mathcal{I}_1} \rangle$ of G , where $\mathcal{W}_1 = \langle \Delta_1, \cdot^{C_1}, \cdot^{P_1} \rangle$ such that $v \in (a^{\mathcal{I}_1})^{C_1}$ and $v \notin (b^{\mathcal{I}_1})^{C_1}$ for some $v \in \Delta_1$. Note that such a model of G exists, because $G \not\models \langle a \text{ subClassOf } b \rangle$. We show that no homomorphism exists from $\mathcal{I}_0$ to $\mathcal{I}_1$. In fact, consider any total function from Δ_0 to Δ_1 such that $\Psi(a^{\mathcal{I}_0}) = a^{\mathcal{I}_1}$, $\Psi(b^{\mathcal{I}_0}) = b^{\mathcal{I}_1}$, $\Psi(\text{subClassOf}^{\mathcal{I}_0}) = \text{subClassOf}^{\mathcal{I}_1}$ and note that while $\langle a^{\mathcal{I}_0}, b^{\mathcal{I}_0} \rangle \in (\text{subClassOf}^{\mathcal{I}_0})^{P_0}$, we have $\langle \Psi(a^{\mathcal{I}_0}), \Psi(b^{\mathcal{I}_0}) \rangle \notin (\text{subClassOf}^{\mathcal{I}_1})^{P_1}$. Therefore, we conclude that Ψ is not a homomorphism from $\mathcal{I}_0$ to $\mathcal{I}_1$.

2. The case of $(a^{\mathcal{I}_0})^{C_0} \neq \emptyset$.

Note that this case can indeed occur, because $G \not\models \langle a \text{ subClassOf } b \rangle$ and it obviously implies that $v \in (a^{\mathcal{I}_0})^{C_0}$ for some $v \in \Delta_0$. Now, consider a model $\mathcal{I}_2 = \langle \mathcal{W}_2, \cdot^{\mathcal{I}_2} \rangle$ of G , where $\mathcal{W}_2 = \langle \Delta_2, \cdot^{C_2}, \cdot^{P_2} \rangle$ such that $(a^{\mathcal{I}_2})^{C_2} = \emptyset$. Note that such a model of G exists, because $G \not\models \langle x \text{ type } a \rangle$, for every $x \in (\mathbb{I} \cup \mathbb{B})$. We show that no homomorphism exists from $\mathcal{I}_0$ to $\mathcal{I}_2$. In fact, consider any total function from Δ_0 to Δ_2 such that $\Psi(a^{\mathcal{I}_0}) = a^{\mathcal{I}_2}$, $\Psi(b^{\mathcal{I}_0}) = b^{\mathcal{I}_2}$, $\Psi(\text{type}^{\mathcal{I}_0}) = \text{type}^{\mathcal{I}_2}$ and note that while $\langle v, a^{\mathcal{I}_0} \rangle \in (\text{type}^{\mathcal{I}_0})^{P_0}$, we have $\langle \Psi(v), \Psi(a^{\mathcal{I}_0}) \rangle \notin (\text{type}^{\mathcal{I}_2})^{P_2}$. Therefore, we conclude that Ψ is not a homomorphism from $\mathcal{I}_0$ to $\mathcal{I}_2$.

So, we have proved that no interpretation of G is a canonical model. $\square$

The previous proposition is a rather surprising result, since RDFS is generally considered a “lightweight” language and such languages typically admit a canonical model, which is exploited for reasoning, e.g. for query entailment. An interesting implication of our result is illustrated by the following example. Consider the RKG $G = \{ \langle a \text{ R } b \rangle, \langle b \text{ R } a \rangle, \langle t \text{ type } b \rangle, \langle a \text{ type Class} \rangle \}$ and the query Q

ASK WHERE $\{ _x \text{ R } _y. _z \text{ type } _y. _x \text{ subClassOf } b \}$

asking whether in every model of G there is a subclass $_x$ of b and a non-empty class $_y$ that are related to each other by R .

If we adopt the standard entailment regime of SPARQL, in order to answer the query we have to look for a binding (homomorphism) that makes the query true in every model of G . By inspecting the triples in G , we realize that, for this purpose, we can limit our attention to two types of bindings: the ones with $\{ _x \rightarrow a, _y \rightarrow b \}$, and the ones with $\{ _x \rightarrow b, _y \rightarrow a \}$. No binding of the first type makes the query true in every model of G , because $\langle a \text{ subClassOf } b \rangle$ does not hold in every model; similarly, no binding of the second type makes the query true, because $\langle _z \text{ type } a \rangle$ does not hold in every model. We then conclude that Q is not entailed by G under the entailment regime of SPARQL. Conversely, according to the RKG Metamodeling Semantics proposed in this paper, we have to check

whether for every model there is a binding that makes the query true. Now, given a model $\mathcal{I}$ of G , the class a is either empty or non-empty in $\mathcal{I}$. In the first case, the binding $\{ _x \rightarrow a^{\mathcal{I}}, _y \rightarrow b^{\mathcal{I}}, _z \rightarrow t^{\mathcal{I}} \}$ makes the query true, whereas in the second case the binding $\{ _x \rightarrow b^{\mathcal{I}}, _y \rightarrow a^{\mathcal{I}}, _z \rightarrow o \}$ makes the query true, where o is any object in $(a^{\mathcal{I}})^C$. We then conclude that Q is entailed by G under the RKG Metamodeling Semantics.

By inspecting the proof of Proposition 1, one can easily realize that the lack of canonical model for G is strongly related to the fact that G does not provide definite information about non-emptiness of the extension of a in all the models of G . To capture this idea of indefiniteness, we introduce the following definitions.

Definition 5. Let G be an RKG and let α be one of its elements (a class or a property). We say that

- α is a populated class or property in G if its extension (as a class or property, respectively) is non-empty in every model of G ;
- α is a bottom class in G if it is a subset of every class in every model of G ;
- α is a bottom property in G if it is a subset of every property in every model of G , and both its range and its domain are subsets of every class in every model of G .

Note that, since RDFS has no form of negation, it does not allow imposing emptiness of a given element. However, if α is non-empty in some of the models of G and empty in others, then a method that most closely resembles modeling α as an element with an empty extension is to make α to be a subset of all other elements (classes or properties, depending of the type of α). This motivates the notions of bottom class and bottom property: a bottom element in G can be safely assumed to be empty in all the models of G .

Definition 6. Let G be an RKG and let α be one of its elements (a class or a property). We say that

- α is a definite class in G if it is a populated or a bottom class in G , and is a definite property if it is a populated or bottom property in G ;
- G is definite if all its classes and properties are definite.

An element or a graph that is not definite is called indefinite.

Observe that in the proof of Proposition 1, class a is indefinite, and so is graph G . Let us analyze the RKG G introduced in Example 1.

Example 3. It is easy to verify that all classes and properties in the RKG G of Example 1 are definite, because they are all populated in G . For example, Person is a definite class because $G \models \langle _ : b \text{ type Person} \rangle$ and teachesTo is a definite property because $G \models \langle _ : b \text{ teachesTo Alice} \rangle$.

We have already seen an example of indefinite RKG that do not admit a canonical model, in the proof of Proposition 1. Actually, in the next section, we will see that definiteness is both a sufficient and a necessary condition for an RKG to admit a canonical model.

5 The chase procedure for RKGs

In this section we define a chase procedure [13] for RKGs and show that it can be a useful tool for query answering over RKGs.

The basic idea of the chase for RKGs is similar to that of the chase in lightweight ontology languages such as $DL\text{-}Lite_{\mathcal{R}}$ [3]: given an RKG G , the procedure consists of a sequence of incremental extensions of G . First, starting from the alphabet $(\mathbb{I}^G \cup \mathbb{B}^G)$ we add to

such alphabet the set of symbols $\mathcal{S} = \{s_e, s'_e, s''_e \mid e \in (\mathbb{I}^G \cup \mathbb{B}^G)\}$ whose role will be clear in the following. Second, we define $Ch^0(G)$ to be the set of triples in G extended with the set of triples listed in Table (a) of Appendix A, which, intuitively, are the triples that are valid, i.e., true in every model of every RKG (e.g., (Resource type Class)).

Then, we proceed by applying suitable rules, called *chase rules*. Specifically, at each step $j \geq 0$, we compute $Ch^{j+1}(G)$ from $Ch^j(G)$ by applying one such rule. We do so following a deterministic strategy that is “fair”, i.e., is such that if at some point a rule is applicable, then it will be eventually applied. The complete list of rules appears in Section A.1 of the Appendix, where, for the sake of conciseness, the symbols of the RDFS vocabulary is expressed by means of abbreviations, i.e., sco (resp., spo, dom, ran) for subClassOf (resp., subPropertyOf, domain, range). They are similar to the inference rules proposed in [11] for the problem of triple entailment in RDFS. There are however, crucial additions because our target problems is query entailment. In particular, we just report two rules here, which, as we will see, play a crucial role. Specifically:

- **rule 22:** if $\langle b \text{ type } a \rangle$ belongs to $Ch^j(G)$ and $\langle s_a \text{ type } a \rangle$ does not belong to $Ch^j(G)$, then $Ch^{j+1}(G)$ is obtained from $Ch^j(G)$ by adding $\langle s_a \text{ type } a \rangle$; and
- **rule 23:** if $\langle a \text{ p } b \rangle$ belongs to $Ch^j(G)$ and $\langle s'_p \text{ p } s''_p \rangle$ does not belong to $Ch^j(G)$, then $Ch^{j+1}(G)$ is obtained from $Ch^j(G)$ by adding $\langle s'_p \text{ p } s''_p \rangle$.

Notice that, while for several formalisms the “chase procedure” might not terminate (making the chase an infinite set), in the case of RKGs, since they do not include existential quantifiers, it always leads to an RKG consisting of a finite set of triples. Thus, assume that no rule is applicable at step j . Then, we call *chase of G* , denoted $Ch(G)$, the RKG $Ch^j(G)$. It is easy to see that $Ch(G)$ can be seen as the structure $CS(G) = \langle \bar{\mathcal{W}}, \bar{\cdot}^{\bar{C}}, \bar{\cdot}^{\bar{P}} \rangle$ defined as follows:

- $\bar{\mathcal{W}} = \langle \bar{\Delta}, \bar{\cdot}^{\bar{C}}, \bar{\cdot}^{\bar{P}} \rangle$ is such that:
 - $\bar{\Delta} = (\mathbb{I}^G \cup \mathbb{B}^G \cup \mathcal{S})$;
 - for every $e \in \bar{\Delta}$, we have that:
 - (i) $e^{\bar{C}}$ is defined for every $e \in \bar{\Delta}$ such that $\langle e \text{ type } \text{Class} \rangle \in Ch(G)$;
 - (ii) $e^{\bar{P}}$ is defined for every $e \in \bar{\Delta}$ such that $\langle e \text{ type } \text{Property} \rangle \in Ch(G)$;
 - (iii) $e^{\bar{P}} = \{(e', e'') \mid \langle e' \text{ e } e'' \rangle \in Ch(G)\}$;
 - (iv) $e^{\bar{C}} = \{e' \mid \langle e' \text{ type } e \rangle \in Ch(G)\}$;
- $\bar{\cdot}^{\bar{P}}$ is a function from $(\mathbb{I}^G \cup \mathbb{B}^G \cup \mathcal{S})$ to $\bar{\Delta}$ such that $e^{\bar{P}} = e$;

Note that in $CS(G)$ blank nodes are directly interpreted by $\bar{\cdot}^{\bar{P}}$ and thus there is no need to look for an assignment in order to check whether the structure $CS(G)$ satisfies a triple of G .

The following proposition shows that $Ch(G)$ enjoys a notable property.

Proposition 2. *Let G be an RKG and let α be any triple over $(\mathbb{I}^G \cup \mathbb{B}^G)$. Then $G \models \alpha$ if and only if $\alpha \in Ch(G)$.*

Proof. (Sketch) The “ $\Leftarrow$ ” direction can be easily proved based on the correctness of the chase rules, i.e., based on the fact that they produce only triples that are entailed by G . As for the “ $\Rightarrow$ ” direction, by construction, if $\alpha \notin Ch(G)$, then $CS(G)$ does not satisfy α . Clearly, if $CS(G)$ is a model of G , then we are done. If $CS(G)$ is

not a model of G , then we prove that we can always modify it in order to obtain a model of G where α is still not satisfied.

More precisely, it can be shown that $CS(G)$ is not a model of G , then it is because it interprets the terms subClassOf, subPropertyOf, domain, and range wrongly. For example, subClassOf may be interpreted in $CS(G)$ as an object whose property extension does not contain a pair (a, b) such that $a^{\bar{C}} \subseteq b^{\bar{C}}$. If this is the case, intuitively, we modify $CS(G)$ so as to include a new object in $a^{\bar{C}}$ and show that this falsifies the subset inclusion between the class extensions of a and b , still falsifying does not satisfy α . $\square$

We are ready to show that definiteness is both a necessary and a sufficient condition for $CS(G)$ to be a canonical model of an RKG G .

Proposition 3. *For an RKG G , $CS(G)$ is a canonical model of G if and only if G is definite.*

Proof. (Sketch) The proof of the “ $\Rightarrow$ ” direction follows straightforwardly from the proof of Proposition 1. As for the “ $\Leftarrow$ ” direction, the proof proceeds by showing that from the definiteness of G follows that $CS(G)$ is a model of G . Also, by induction on the number k of rules applied to $Ch^0(G) = G$ to obtain the portion $Ch^k(G)$ of $Ch(G)$, it shows that for every model $\mathcal{J}$ of G , there exists an RKG homomorphism from $\bar{\mathcal{I}}$ to $\mathcal{J}$. Note that the proof is very similar to the one used in [4] for *DL-Lite_R* ontologies. The only difference concerns the cases where at step k either rule 22 or rule 23 is applied. Intuitively, in both cases a new triple is introduced, which implies that an object denoted by a blank node not occurring anywhere else in $Ch^k(G)$ exists that belongs to the extension of a class, already known to be non empty (or respectively that a pair of objects denoted by a pair of blank nodes not occurring anywhere else in $Ch^k(G)$ exist, that belongs to the extension of a property, already known to be non empty). Thus, clearly, if an RKG homomorphism from $Ch^k(G)$ seen as a structure to $\mathcal{J}$ exists, then it can be trivially extended to the newly introduced objects so as to obtain an RKG homomorphism from $Ch^{k+1}(G)$ to $\mathcal{J}$. $\square$

An easy corollary of the above proposition is that an RKG admits a canonical model if and only if it is definite. Also, an algorithm can be easily derived for query entailment in definite RKGs, based on constructing $CS(G)$ and then checking for a query homomorphism from the query Q to $CS(G)$. The following is an easy consequence of this property.

Proposition 4. *Query entailment in definite RKGs is NP-complete in combined complexity.*

We finally show that $CS(G)$ can be exploited to check in polynomial time whether any symbol is definite.

Proposition 5. *An element $a \in (\mathbb{I}^G \cup \mathbb{B}^G)$ is indefinite in G if at least one of the following holds:*

- $\langle a \text{ type } \text{Class} \rangle \in Ch(G)$, for every b , $\langle b \text{ type } a \rangle \notin Ch(G)$ and there exists c such that $\langle c \text{ type } \text{Class} \rangle \in Ch(G)$ and $\langle a \text{ subClassOf } c \rangle \notin Ch(G)$.
- $\langle a \text{ type } \text{Property} \rangle \in Ch(G)$, for every b and every c , $\langle b \text{ a } c \rangle \notin Ch(G)$, there exists p such that $\langle p \text{ type } \text{Property} \rangle \in Ch(G)$ and $\langle a \text{ subPropertyOf } p \rangle \notin Ch(G)$, and at least one of the following holds:
 - there exists d such that $\langle d \text{ type } \text{Class} \rangle \in Ch(G)$, $\langle a \text{ domain } d \rangle \notin Ch(G)$;

- there exists d such that $\langle d \text{ type class} \rangle \in Ch(G)$,
 $\langle a \text{ range } d \rangle \notin Ch(G)$.

Proof. The proof immediately follows from the definition of definiteness and Proposition 2. $\square$

6 Query entailment for the general case

With all the notions in place, we are now ready to present an algorithm for solving the problem of query entailment in the general case, i.e., for those cases where the RKG the query is posed against is indefinite. We remind the reader that, as a consequence of Proposition 1, for an indefinite RKG, no model can be used as a semantical representative of every model, and we are therefore forced to address the task of query entailment by means of reasoning by cases.

The algorithm we present is based on the notion of completion of an RKG G . Intuitively, a completion of G is obtained from G by removing the indefiniteness of the elements (classes and properties) of G . This is done by choosing a subset φ of the whole set Φ of indefinite elements of G , making them nonempty in the completion of G (called G^φ) and making each of the other indefinite elements E (those in $\Phi \setminus \varphi$) a subset of all the elements (class or properties, depending on the type of E) in G^φ , so as to simulate the emptiness of E . More formally,

$$G^\varphi = G \cup \{ \langle s_E \text{ type } E \rangle \mid E \in \varphi \text{ is a class} \} \cup \\ \{ \langle s_E^1 E s_E^2 \rangle \mid E \in \varphi \text{ is a property} \} \cup \\ \{ \langle E \text{ subClassOf } F \rangle \mid E \in \Phi \setminus \varphi \text{ is a class and} \\ F \text{ is a class in } G \} \cup \\ \{ \langle E \text{ subPropertyOf } F \rangle, \langle E \text{ domain } C \rangle, \langle E \text{ range } C \rangle \mid \\ E \in \Phi \setminus \varphi \text{ is a property,} \\ F \text{ is a property in } G \text{ and} \\ C \text{ is a class in } G \}$$

where the various s_E, s_E^1, s_E^2 are individuals not appearing in G .

The algorithm for query entailment can then be formulated as follows. Given an RKG G and a conjunctive query Q , do the following:

for each subset φ of Φ :

if $G^\varphi \models Q$ **then return false;**

return true

The algorithm clearly halts for every input, and its correctness is shown in the following theorem.

Theorem 1. *Let G be an RKG, let Φ be the set of indefinite elements in G , and Q a conjunctive query. $G \models Q$ if and only if $G^\varphi \models Q$ for every subset φ of Φ .*

Proof. (Sketch) The proof is based on showing that $Mod(G) = \bigcup_{\varphi \subseteq \Phi} Mod(G^\varphi)$. Since for every $\varphi \subseteq \Phi$, every model of G^φ is clearly also a model of G , one direction is obvious. For the other direction, observe that in every model of G there will be a subset φ of Φ such that all the elements of φ are nonempty, while each element E in $\Phi \setminus \varphi$ is empty, and therefore E is a subset of all the elements (class or properties, depending of the type of E) of G . $\square$

From the correctness of the above algorithm it immediately follows that query entailment in our setting can be done in Π_2^P with respect to the size of the entire input. More precisely, we could show that the problem is in coNP with respect to the size of the RKG and in NP with respect to the size of the query.

We now prove that the problem of query entailment for RKGs is Π_2^P -hard with respect to the size of both the graph and the query. To do so, we rely on a reduction from the satisfiability problem for 2-QBF, i.e., Quantified Boolean Formulas of the form $\forall X \exists Y. \phi(X, Y)$, where $\phi(X, Y)$ is a 3-CNF formula consisting of k clauses and such

that $X = \{x_1, \dots, x_n\}$ and $Y = \{y_1, \dots, y_m\}$ represent the set of universally quantified variables and existentially quantified variables, respectively.

Due to space limitations, here we provide only an intuition of the reduction, and defer the full details to Appendix C.

Initially, a 2-QBF formula F is transformed in a new formula F' whose size is linear with respect to the size of F , and such that F' is satisfiable if and only if F is satisfiable. Starting from the formula F' , it is possible to build an RKG $G^{F'}$ in such a way that its axioms are meant to encode, for each clause of F' , all possible truth assignments in such a way that each existential variable of F' is assigned a truth value, while all universally quantified variables of F' are kept undefined. The graph includes individuals introduced to identify existentially quantified variables by means of their position within the clauses. Moreover, variables assigned to false are indicated by means of a connection with an empty class E , while variables assigned to true by means of a connection to a non-empty class D . Note that, to reflect the emptiness of class E , the graph must include axioms assessing its subclass relationship with respect to every class of $G^{F'}$, and the empty class E is also used to indicate emptiness of other classes by making them subclasses of E . Universally quantified variables x_j being kept undefined is reflected on the presence of corresponding classes x_j in $G^{F'}$ which are indefinite, thus making $G^{F'}$ an indefinite RKG. By means of the completion algorithm described for answering queries, it is possible to compute the complete set of definite RKGs derived from $G^{F'}$. Each such definite RKG is meant to encode truth assignments for the universally quantified variables of F' by getting rid of the indefiniteness that characterize them, i.e., by either including axioms of the form $\langle x \text{ subClassOf } E \rangle$ for variables to be assigned to false, or violating such axiom. Intuitively, the formula F' is satisfiable if, for every definite RKG $G_d^{F'}$ derived from $G^{F'}$, there exists a subset of the axioms of $G_d^{F'}$ that encodes an assignment for the variables of F' satisfying all its clauses.

To check the existence of such assignment, we introduce a Boolean conjunctive query $Q^{F'} \leftarrow \phi_1 \wedge \phi_2$, where ϕ_1 is used to check for the existence of atoms in the derived definite RKG encoding an assignment to the variables of a clause c_i in such a way that the assignment makes the clause true, while ϕ_2 is used to check the existence of atoms in the derived definite graph encoding an assignment ensuring the consistency of assignments.

Theorem 2. *F is satisfiable if and only if $G^{F'} \models Q^{F'}$*

The proof of the above Theorem is presented in Appendix D.

Since $G^{F'}$ and $Q^{F'}$ are polynomial in size with respect to the size of the formula F , and considering the results from Theorem 2, we can derive a matching lowerbound of Π_2^P -hard with respect to the size of the entire input for the problem of query entailment over RKGs.

Theorem 3. *Query entailment in RKGs is Π_2^P -complete in combined complexity.*

7 Conclusions

We have proposed a pure logic approach to what probably represents the simplest form of Knowledge Graphs, namely the RDFS Knowledge Graphs. We have shown the surprising result that under the Metamodeling Semantics, which is grounded in logic, the complexity of conjunctive query answering jumps to Π_2^P , due to a certain form of incompleteness in the knowledge expressed in the graph. We have also shown that, without such form of incompleteness, query answering remains in polynomial time. While in this paper we concentrated

on combined complexity, it would be interesting to investigate other complexity measures, e.g., *ontology complexity* (where the query Q is fixed and only the RKG is part of the input), and *query complexity* (where the RKG is fixed and the input consists solely of the query). In particular, we have already studied ontology complexity, but omitted it here due to space limitations.

There are several aspects for continuing our work. We mention two of them here. We are investigating a new formalization of RKGs based on epistemic logic, in order to provide users with alternative interpretations for query semantics, enabling the integration of features from different semantic paradigms. Also, we are addressing the issue of increasing the expressive power of the framework, by extending both the knowledge graph language and the query language with basic forms of negation, inspired by [14, 8, 9].

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A Appendix

A.1 Chase rules

The complete list of rules is specified in Figure 2b, where we denote by a, b, c, p, q, r symbols in $(\mathbb{I}^G \cup \mathbb{B}^G)$, and by s_a, s'_p, s''_p symbols in $\mathcal{S}$.

For each rule, we report the rule number n , the set of axioms P_n and the set of axioms A_n such that the rule n is applicable at step j if $P_n \subseteq Ch^j(G)$ and $A_n \not\subseteq Ch^j(G)$. Also, by applying the rule n at step j , $Ch^{j+1}(G)$ is obtained from $Ch^j(G)$ by uniting A_n .

A.2 Proof of Prop. 2

$\Leftarrow$: Suppose that $\alpha = \langle a \ p \ b \rangle \in Ch(G)$. Then either $\alpha \in G$ or it was added by applying one of the chase rules 1 to 21 (because α comprises only symbols in $(\mathbb{I}^G \cup \mathbb{B}^G)$ and thus it cannot be added though the application of rules 22 and 23). Hence, the proof immediately follows from the correctness of the RDFS entailment rules.

$\Rightarrow$: Suppose that $\alpha = \langle a \ p \ b \rangle \notin Ch(G)$. It is easy to see that, by construction, if $\alpha \notin Ch(G)$, then $CS(G)$ does not satisfy α . Now, clearly, if $CS(G)$ is a model of G , then we are done. Hence, we will consider the case of $CS(G)$ not being a model of G , and we will show how we can modify it, in order to obtain a model of G . Then, we conclude the proof by showing that such model does not satisfy α .

We first suppose that $CS(G)$ is not an interpretation of G because it does not assign to some symbol in $\mathbb{R}$ objects satisfying the requirements established by the definition of RKG interpretation. In fact, we next show that $CS(G)$ satisfies all requirements expressed for Resource, Class, Property, and type, while it may not satisfy requirements for subClassOf, subPropertyOf, domain, and range.

Let us start by considering Resource and assume by contradiction that $(Resource^{\bar{I}})^C \neq \bar{\Delta}$. Clearly, $(Resource^{\bar{I}})^C \subseteq \bar{\Delta}$. Hence, the only possibility is that there exists $o \in \bar{\Delta}$ such that $o \notin (Resource^{\bar{I}})^C$. But, if $o \in \bar{\Delta}$, then o has to occur at least once either in subject, object, or predicate position. If o occurs in subject or object position in G , then, eventually, rules 9 and 10 are applied, which implies that $\langle e \ \text{type} \ \text{Resource} \rangle$ belongs to $Ch(G)$. Thus, by construction of $can(G)$, $o \in (Resource^{\bar{I}})^C$. Similarly, if e occurs in predicate position in G , then, eventually, rule 11 is applied, which implies that $\langle o \ \text{type} \ \text{Property} \rangle$ belongs to $Ch(G)$. Also, since $\langle \text{type} \ \text{domain} \ \text{Resource} \rangle$ is a valid triple, rule 4 is eventually applied and, as a matter of fact, $\langle o \ \text{type} \ \text{Resource} \rangle$ belongs to $Ch(G)$, as well. Hence, for every $o \in \mathbb{I}^G$, we have that $o^{\bar{I}} = o \in (Resource^{\bar{I}})^C$, which contradicts our assumption. Let us now consider Class. Clearly, $(Class^{\bar{I}})^C$ is defined because by construction $Ch^0(G)$ contains the valid triple $\langle \text{Resource} \ \text{type} \ \text{Class} \rangle$. We still need to show that $(Class^{\bar{I}})^C = \{o \in \bar{\Delta} \mid (o^{\bar{I}})^C \text{ is defined}\}$. By construction of $CS(G)$, if $o \in (Class^{\bar{I}})^C$, by (iv) it must hold that $\langle o \ \text{type} \ \text{Class} \rangle$ belongs to $Ch(G)$. But then, by (i), $(o^{\bar{I}})^C$ is defined. Conversely, let $o \in \bar{\Delta}$ be such that $(o^{\bar{I}})^C$ is defined. Then, since $\langle o \ \text{type} \ \text{Class} \rangle$ belongs to $Ch(G)$, by (iv), we have that $o \in (Class^{\bar{I}})^C$. It is easy to see we can proceed analogously when considering Property and type.

Let us now consider subClassOf. By construction, if $(c_1, c_2) \in (subClassOf^{\bar{I}})^P$, then $c_1^C \subseteq c_2^C$. However, the converse may not be true and similar statements hold for subPropertyOf, domain, and range. Hence, without loss of generality, let us suppose that $CS(G)$ is not an interpretation because of the following:

$\langle \text{domain} \ \text{domain} \ \text{Property} \rangle$
$\langle \text{range} \ \text{domain} \ \text{Property} \rangle$
$\langle \text{subPropertyOf} \ \text{domain} \ \text{Property} \rangle$
$\langle \text{subClassOf} \ \text{domain} \ \text{Class} \rangle$
$\langle \text{type} \ \text{range} \ \text{Class} \rangle$
$\langle \text{domain} \ \text{range} \ \text{Class} \rangle$
$\langle \text{range} \ \text{range} \ \text{Class} \rangle$
$\langle \text{subPropertyOf} \ \text{range} \ \text{Property} \rangle$
$\langle \text{subClassOf} \ \text{range} \ \text{Class} \rangle$
$\langle \text{type} \ \text{type} \ \text{Property} \rangle$
$\langle \text{type} \ \text{domain} \ \text{Resource} \rangle$
$\langle \text{Resource} \ \text{type} \ \text{Class} \rangle$

(a). Valid triples

n	P_n	A_n
1	$\langle c \ \text{type} \ a \rangle, \langle a \ \text{sco} \ b \rangle$	$\langle c \ \text{type} \ b \rangle$
2	$\langle a \ p \ b \rangle, \langle p \ \text{spo} \ q \rangle$	$\langle a \ q \ b \rangle$
3	$\langle a \ \text{sco} \ b \rangle, \langle b \ \text{sco} \ c \rangle$	$\langle a \ \text{sco} \ c \rangle$
4	$\langle p \ \text{dom} \ c \rangle, \langle a \ p \ b \rangle$	$\langle a \ \text{type} \ c \rangle$
5	$\langle p \ \text{ran} \ c \rangle, \langle a \ p \ b \rangle$	$\langle b \ \text{type} \ c \rangle$
6	$\langle p \ \text{type} \ \text{Property} \rangle$	$\langle p \ \text{spo} \ p \rangle$
7	$\langle c \ \text{type} \ \text{Class} \rangle$	$\langle c \ \text{sco} \ c \rangle$
8	$\langle p \ \text{spo} \ q \rangle, \langle q \ \text{spo} \ r \rangle$	$\langle p \ \text{spo} \ r \rangle$
9	$\langle a \ p \ b \rangle$	$\langle a \ \text{type} \ \text{Resource} \rangle$
10	$\langle a \ p \ b \rangle$	$\langle b \ \text{type} \ \text{Resource} \rangle$
11	$\langle a \ p \ b \rangle$	$\langle p \ \text{type} \ \text{Property} \rangle$
12	$\langle a \ \text{type} \ \text{Class} \rangle$	$\langle a \ \text{sco} \ \text{Resource} \rangle$
13	$\langle p \ \text{dom} \ b \rangle, \langle b \ \text{sco} \ c \rangle$	$\langle p \ \text{dom} \ c \rangle$
14	$\langle p \ \text{range} \ b \rangle, \langle b \ \text{sco} \ c \rangle$	$\langle p \ \text{ran} \ c \rangle$
15	$\langle q \ \text{dom} \ c \rangle, \langle p \ \text{spo} \ q \rangle$	$\langle p \ \text{dom} \ c \rangle$
16	$\langle q \ \text{ran} \ c \rangle, \langle p \ \text{spo} \ q \rangle$	$\langle p \ \text{ran} \ c \rangle$
17	$\langle \text{type} \ \text{spo} \ p \rangle, \langle p \ \text{dom} \ a \rangle$	$\langle \text{Resource} \ \text{sco} \ a \rangle$
18	$\langle \text{sco} \ \text{spo} \ p \rangle, \langle p \ \text{dom} \ a \rangle$	$\langle \text{Class} \ \text{sco} \ a \rangle$
19	$\langle \text{sco} \ \text{spo} \ p \rangle, \langle p \ \text{range} \ a \rangle$	$\langle \text{Class} \ \text{sco} \ a \rangle$
20	$\langle \text{spo} \ \text{spo} \ p \rangle, \langle p \ \text{dom} \ a \rangle$	$\langle \text{Property} \ \text{sco} \ a \rangle$
21	$\langle \text{spo} \ \text{spo} \ p \rangle, \langle p \ \text{ran} \ a \rangle$	$\langle \text{Property} \ \text{sco} \ a \rangle$
22	$\langle b \ \text{type} \ a \rangle$	$\langle s_a \ \text{type} \ a \rangle$
23	$\langle a \ p \ b \rangle$	$\langle s'_p \ p \ s''_p \rangle$

(b). Chase rules for RKGs

Figure 2: Chase procedure

- there exist c_1 and c_2 such that $c_1^{\bar{C}} \subseteq c_2^{\bar{C}}$ and $(c_1, c_2) \notin (\text{subClassOf}^{\bar{I}})^{\bar{P}}$;
- there exist p_1 and p_2 such that $p_1^{\bar{P}} \subseteq p_2^{\bar{P}}$ and $(p_1, p_2) \notin (\text{subPropertyOf}^{\bar{I}})^{\bar{P}}$;
- there exist p_1 and c such that $\{o_1 \mid \exists o_2(o_1, o_2) \in p_1^{\bar{P}}\} \subseteq c^{\bar{C}}$ and $(p_1, c) \notin (\text{domain}^{\bar{I}})^{\bar{P}}$;
- there exist p_1 and c such that $\{o_1 \mid \exists o_2(o_2, o_1) \in p_1^{\bar{P}}\} \subseteq c^{\bar{C}}$ and $(p_1, c) \notin (\text{range}^{\bar{I}})^{\bar{P}}$.

It is easy to see that the above properties show that $CS(G)$ does not satisfy the requirements for subClassOf , subPropertyOf , domain , and range , respectively.

We next show how to define a structure M' that is a model of G . In order to do so, let us define G' by first extending $Ch(G)$ with the triples $N = \{\langle s_{c_1} \text{ type } c_1 \rangle, \langle s_{p_1} p_1 s_{p_1}' \rangle\}$, and then by repeatedly applying to $(Ch(G) \cup N)$ all applicable chase rules. Clearly, the above procedure terminates. Also, by proceeding in the same way we did to define $CS(G)$ based on $Ch(G)$, we next define $M' = \langle \bar{W}', \cdot^{\bar{I}'} \rangle$, with $\bar{W}' = \langle \bar{\Delta}, \cdot^{\bar{C}'}, \cdot^{\bar{P}'} \rangle$, based on G' .

Let us now show that M' is a model of G . We need to show that M' is an interpretation of G . Following the same line of reasoning as for $CS(G)$, M' satisfies all requirements expressed for Resource , Class , Property , and type by construction. Hence, it remains to show that M' satisfies all requirements expressed for the remaining symbols of $\mathbb{R}$, and in particular those that were not satisfied by $CS(G)$.

Let us consider subClassOf . We assumed that $c_1^{\bar{C}} \subseteq c_2^{\bar{C}}$ and $(c_1, c_2) \notin (\text{subClassOf}^{\bar{I}})^{\bar{P}}$. It follows that $c_1 \neq c_2$ (otherwise, by rule 7 we would get a contradiction) and that $c_2 \neq \text{Resource}$ (otherwise, by rule 12 we would get a contradiction). We next show that $s_{c_1} \notin c_1^{\bar{C}}$. We thus suppose, by contradiction, that $s_{c_1} \in c_1^{\bar{C}}$. Since $c_1^{\bar{C}} \subseteq c_2^{\bar{C}}$, we have that, $s_{c_1} \in c_2^{\bar{C}}$, which implies that $\langle s_{c_1} \text{ type } c_2 \rangle$ belongs to $Ch(G)$, where $c_2 \neq \text{Resource}$. But then the latter was added by applying a chase rule. In fact, by inspecting the list of chase rules, we can see that the only rules that could add such triple are rules 1 or 4. We next show that in fact none of these rules can be applied:

- in order to apply rule 1, there must exist c' such that $\langle c' \text{ subClassOf } c_2 \rangle$ and $\langle s_{c_1} \text{ type } c' \rangle$ belong to $Ch(G)$; clearly, if $c' = c_1$, we get a contradiction; moreover, since $Ch(G)$ contains the set of triples that is obtained by repeatedly applying rule 3 and rule 7 to the set of triples with predicate subClassOf occurring in G , (i.e., $\text{subClassOf}^{\bar{I}})^{\bar{P}}$ is transitive), c' is such that there exist no triple $\langle c_1 \text{ subClassOf } c' \rangle$ in $Ch(G)$; hence, the only remaining possible case, is that $c' = \text{Resource}$ and that rule 17 has been applied, but again this leads to a contradiction, because, as already discussed, $\langle c_1 \text{ subClassOf } \text{Resource} \rangle$ in $Ch(G)$, and hence, by rule 3, $\langle c_1 \text{ subClassOf } c_2 \rangle$ belongs to $Ch(G)$, which contradicts our assumption;
- in order to apply rule 4, there must exist p and c such that $\langle p \text{ domain } c_2 \rangle$ and $\langle s_{c_1} p c \rangle$ belong to $Ch(G)$; but then, since $\langle s_{c_1} \text{ type } c_1 \rangle$, this can happen if either $p = \text{type}$ or $\langle \text{type subPropertyOf } p \rangle$ belongs to $Ch(G)$; in both cases, by rule 17, we have that $\langle \text{Resource subClassOf } c_2 \rangle$ belongs to $Ch(G)$ and hence, since $\langle c_1 \text{ subClassOf } c_2 \rangle$, we get a contradiction.

Hence, $s_{c_1} \notin c_1^{\bar{C}}$, while, by construction of M' , $s_{c_1} \in c_1^{\bar{C}'}$. Clearly, following the same line of reasoning as before, we can conclude that $s_{c_1} \notin c_2^{\bar{C}'}$.

On the other hand, it is easy to see that the addition of the

triples in N does not trigger the introduction of triples with predicate subClassOf . As a matter of fact, we have that for every c_1, c_2 , $c_1^{\bar{C}'} \subseteq c_2^{\bar{C}'}$ if and only if $(c_1, c_2) \in (\text{subClassOf}^{\bar{I}'})^{\bar{P}'}$. Hence, M' interprets subClassOf as required by the definition of RKG interpretation.

By following a similar yet easier line of reasoning, we can show that M' interprets subPropertyOf , domain , and range as required by the definition of RKG interpretation. Also, since it is immediate to verify that, by construction, M' satisfies all triples of G , M' is a model of G .

In order to conclude the proof we next show that for every triple $\alpha = \langle a p b \rangle$ over $(\mathbb{I}^G \cup \mathbb{B}^G)$, if $\alpha \notin Ch(G)$, then $M' \not\models \alpha$, thus showing that $G \not\models \alpha$. By construction, if $\alpha \notin G'$, then $M' \not\models \alpha$. Hence, suppose by contradiction that $\alpha \in G'$. Since G' is an extension of $Ch(G)$, the only possibility is that α is either one of the triples of N , or it was introduced by repeatedly applying the chase rules to $(Ch(G) \cup N)$. Note however that α is a triple not containing any symbol in S . By inspecting the chase rules, one can easily verify that this leads to a contradiction because the only chase rules that are applicable to $(Ch(G) \cup N)$, are Rule 1 and Rule 4, and both lead to introduce only triples of the same form of those in N and containing symbols in S . $\square$

B Complete proof of Proposition3

$\Leftarrow$: The proof follows from the proof of Proposition 1.

$\Rightarrow$: By definition, in order for $CS(G)$ to be a canonical model:

1. $CS(G)$ must be a model of G ;
2. for every model $\mathcal{J}$ of G , there exists an RKG homomorphism from $\bar{\mathcal{I}}$ to $\mathcal{J}$.

Recall the proof of Proposition 2. The only conditions preventing $CS(G)$ from being an interpretation are the following:

- there exist c_1 and c_2 such that $c_1^{\bar{C}} \subseteq c_2^{\bar{C}}$ and $(c_1, c_2) \notin (\text{subClassOf}^{\bar{I}})^{\bar{P}}$;
- there exist p_1 and p_2 such that $p_1^{\bar{P}} \subseteq p_2^{\bar{P}}$ and $(p_1, p_2) \notin (\text{subPropertyOf}^{\bar{I}})^{\bar{P}}$;
- there exist p_1 and c such that $\{o_1 \mid \exists o_2(o_1, o_2) \in p_1^{\bar{P}}\} \subseteq c^{\bar{C}}$ and $(p_1, c) \notin (\text{domain}^{\bar{I}})^{\bar{P}}$;
- there exist p_1 and c such that $\{o_1 \mid \exists o_2(o_2, o_1) \in p_1^{\bar{P}}\} \subseteq c^{\bar{C}}$ and $(p_1, c) \notin (\text{range}^{\bar{I}})^{\bar{P}}$.

Hence, suppose by contradiction the first condition is satisfied. Then, since G is empty-definite, by definition, all symbols in $(\mathbb{I}^G \cup \mathbb{B}^G)$ are definite. Hence, either there exists b such that $G \models \langle b \text{ type } c_1 \rangle$, or for every d such that $G \models \langle d \text{ type } \text{Class} \rangle$, $G \models \langle c_1 \text{ subClassOf } d \rangle$. In the former case, by construction of $CS(G)$, $s_{c_1} \in c_1^{\bar{C}}$. Clearly, based on Proposition 2, both cases lead to contradiction. An analogous proof can be followed for the other conditions, which shows that $CS(G)$ is an interpretation of G .

Let us now show that for every model $\mathcal{J}$ of G , there exists an RKG homomorphism from $\bar{\mathcal{I}}$ to $\mathcal{J}$. Let $\mathcal{J} = \langle \mathcal{W}, \cdot^{\bar{\mathcal{I}}} \rangle$ be a model of G , such that $\mathcal{W} = \langle \bar{\Delta}, \cdot^{\bar{\mathcal{C}}}, \cdot^{\bar{\mathcal{P}}} \rangle$, $\nu^{\bar{\mathcal{I}}}$ is an assignment function coherent with $\bar{\mathcal{I}}$ such that $\bar{\mathcal{I}}$ satisfies all triples of G under $\nu^{\bar{\mathcal{I}}}$, and Ψ a function from $\bar{\Delta} = (\mathbb{I}^G \cup \mathbb{B}^G \cup S)$ to Δ such that for every $e \in \mathbb{I}^G$, $\Psi(e) = e^{\bar{\mathcal{I}}}$, for every $e \in \mathbb{B}^G$, $\Psi(e) = \nu^{\bar{\mathcal{I}}}(e)$ and for every $e \in S$, $\Psi(e) = \text{Resource}$. In fact, in the following, while showing that Ψ is an RKG homomorphism from $\bar{\mathcal{I}}$ to $\mathcal{J}$, we will possibly that Ψ , obtained by possibly modify $\Psi(e)$ for some $e \in S$.

Before proceeding, let the k -portion of $CS(G)$, denoted $CS_k(G) = \langle \bar{W}_k, \bar{\mathcal{I}} \rangle$, be the structure based on $Ch^k(G) = \bigcup_{i \leq 0 \leq k} Ch^i(G)$, in the sense that its definition differs from that of $CS(G)$ only because it uses $Ch^k(G)$ instead of $Ch(G)$, thus obtaining $\bar{W}_k = \langle \bar{\Delta}_k, \bar{C}_k, \bar{P}_k \rangle$. In order to show that Ψ is an RKG homomorphism from $\bar{\mathcal{I}}$ to $\mathcal{J}$, we can now proceed by induction on the number k where k is the number of rules that were applied to $Ch^0(G) = G$ to obtain $Ch^k(G)$. As for the base step, we have to show that for every $o_1, o_2, o_3 \in \bar{\Delta}$ such that $\langle o_2, o_3 \rangle \in o_1^{P_0}$, $\langle \Psi(o_2), \Psi(o_3) \rangle \in \Psi(o_1)^P$. Since $\langle o_2, o_3 \rangle \in o_1^{P_0}$, by construction of $CS_0(G)$, we have that $\langle o_2 \ o_1 \ o_3 \rangle$ belongs to $Ch^0(G)$. Hence, it is either contained in G or it is one of the valid triples of Fig. 2a. Clearly, in both cases, such a triple has to be satisfied also by $\mathcal{J}$ and hence, given the way Ψ is defined, it follows that $\langle \Psi(o_2), \Psi(o_3) \rangle \in \Psi(o_1)^P$. As for the inductive step, let k be the number of rules that led to $Ch^k(G)$ and suppose that for every $o_1, o_2, o_3 \in \bar{\Delta}$ such that $\langle o_2, o_3 \rangle \in o_1^{P_k}$, $\langle \Psi(o_2), \Psi(o_3) \rangle \in \Psi(o_1)^P$. Since this proof is very similar to the one used in [4] for *DL-Lite_R* ontologies, and is based on the correctness of the RDFS entailment rules, in the following, we consider only the case where at step k is applied either rule 22 or rule 23, which are also the only cases where Ψ is modified. Let us consider the first case. Since the rule is applicable, it must be the case that $\langle b \text{ type } c \rangle$ belongs to $Ch^k(G)$ and $\langle s_c \text{ type } c \rangle$ does not. Then, the latter triple is added to $Ch^{k+1}(G)$ which implies that $\langle s_c, c \rangle \in \text{type}^{P_{k+1}}$. Also, we modify Ψ by setting $\Psi(s_c) = \Psi(b)$. By inductive hypothesis, we only need to show that $\langle \Psi(s_c), \Psi(c) \rangle \in \Psi(\text{type})^P$. However, it is immediate to verify that this follows from the fact that $\langle b \text{ type } c \rangle$ belongs to $Ch^k(G)$. Also, the case of rule 23 is analogous, which concludes the proof. $\square$

C Reduction from Satisfiability of 2-QBF Formulas

Starting from a 2-QBF formula F we can obtain a formula F' as follows. For each variable z of F :

- we introduce a new variable $\bar{z}$;
- we substitute each negative literal in F with $\bar{z}$;
- we add every $\bar{z}$ to Y ;
- we add clauses $c_{z,1}, c_{z,2}$ to F whose role is to sanction that exactly one between z and $\bar{z}$ is true, having $c_{z,1} = (z \vee \bar{z})$ and $c_{z,2} = (\neg z \vee \neg \bar{z})$. We call such new clauses *auxiliary clauses*, while we call *original clauses* the ones already present in F .

The size of the derived formula F' is linear with respect to the size of F . Moreover, F' is satisfiable if and only if F is satisfiable. In the next steps, we will concentrate on the formula F' for the reduction.

Starting from F' we derive an RKG $G^{F'}$ as follows. First of all, we specify the set of classes, properties and individuals $G^{F'}$ is constituted by.

- Classes include $\{D, E\}$ and a class x_j for each $x_j \in X$;
- Individuals include $\{a\}$; for each original clause c_i , $G^{F'}$ also include individuals c_i, t_{l_1, l_2, l_3}^i , and $v_{l_1, l_2, l_3}^{i, h}$ with $l_1, l_2, l_3 \in \{0, 1\}$, $h \in \{1, 2, 3\}$ and such that at least l_h 's is equal to 1; for each variable z in F' , $G^{F'}$ includes individuals $c_{z,1}, t_{l_1, l_2}^z$, and $v_{l_1, l_2}^{z, h}$ with $l_1, l_2 \in \{0, 1\}$, $h \in \{1, 2\}$, and such that exactly one l_h is equal to 1. Finally, the graph also includes individuals x_j for each $x_j \in X$;
- Properties include $\{\text{In}, \text{Has}, \text{Val}, \text{type}, \text{subClassOf}\}$.

The reduction is based on the idea that the triples in $G^{F'}$ are meant to encode, for each clause of F' , all possible assignments in such a way that each $y \in Y$ is assigned a truth value, while all $x \in X$ are kept undefined. The individuals introduced allow to identify existentially quantified variables by means of their position within the clauses. Moreover, variables assigned to false are indicated by means of a connection with the empty class E , while variables assigned to true by means of a connection to the class D . Note that, to reflect the emptiness of class E , the graph will include axioms assessing its subclass relationship with respect to every class of $G^{F'}$, and it will also be used to indicate emptiness of other classes by making them subclasses of E . Universally quantified variables being kept undefined is reflected on the presence of corresponding classes which are indefinite in $G^{F'}$, thus making $G^{F'}$ an indefinite RKG. Each definite RKG derived from $G^{F'}$ is meant to encode truth assignments for the universally quantified variables by getting rid of the indefiniteness that characterize them, i.e., by either including axioms of the form $\langle x \text{ subClassOf } E \rangle$ for variables to be assigned to false, or violating such axiom.

Then, we can build $G^{F'}$ by adding the following axioms to it:

- $\langle E \text{ subClassOf } C \rangle$, for every class C in $G^{F'}$;
- $\langle a \text{ type } D \rangle$;
- $\langle c_i \text{ In } t_{l_1, l_2, l_3}^i \rangle$ for every clause c_i and for every t_{l_1, l_2, l_3}^i ;
- $\langle c_{z,1} \text{ In } t_{l_1, l_2}^z \rangle$, for each variable z and for every t_{l_1, l_2}^z ;
- for each axiom $\langle c_i \text{ In } t_{l_1, l_2, l_3}^i \rangle$ and for each $h \in \{1, 2, 3\}$ with z_h the h -th variable of c_i , add axiom $\langle t_{l_1, l_2, l_3}^i \text{ Has } z_h \rangle$ if z_h is universally quantified, add axiom $\langle t_{l_1, l_2, l_3}^i \text{ Has } v_{l_1, l_2, l_3}^{i, h} \rangle$ otherwise;
- for each axiom $\langle c_{z,1} \text{ In } t_{l_1, l_2}^z \rangle$, add axiom $\langle t_{l_1, l_2}^z \text{ Has } v_{l_1, l_2}^{z, 2} \rangle$; moreover, add axiom $\langle t_{l_1, l_2}^z \text{ Has } z \rangle$ if z is universally quantified, add axiom $\langle t_{l_1, l_2}^z \text{ Has } v_{l_1, l_2}^{z, 1} \rangle$ otherwise;
- for each $\langle t_{l_1, l_2, l_3}^i \text{ Has } v_{l_1, l_2, l_3}^{i, h} \rangle$, add axiom $\langle v_{l_1, l_2, l_3}^{i, h} \text{ Val } D \rangle$ if $l_h = 1$, axiom $\langle v_{l_1, l_2, l_3}^{i, h} \text{ Val } E \rangle$ otherwise;
- for each $\langle t_{l_1, l_2}^z \text{ Has } v_{l_1, l_2}^{z, 1} \rangle$, add axiom $\langle v_{l_1, l_2}^{z, h} \text{ Val } D \rangle$ if $l_h = 1$, axiom $\langle v_{l_1, l_2}^{z, h} \text{ Val } E \rangle$ otherwise;
- for each universally quantified variable x_j in F' , axiom $\langle x_j \text{ Val } x_j \rangle$

The formula F' is satisfiable if, for every definite RKG $G_d^{F'}$ derived from $G^{F'}$, there exists a subset of the triples of $G_d^{F'}$ that encodes an assignment for the variables of F' satisfying all its clauses.

Such assignments can be checked through a conjunctive query $Q^{F'} \leftarrow \phi_1 \wedge \phi_2$, such that ϕ_1 is constituted by the atoms $\langle c_i \text{ In } w_i \rangle, \langle w_i \text{ Has } s_{i,1} \rangle, \langle s_{i,1} \text{ Val } a_{i,1} \rangle, \langle w_i \text{ Has } s_{i,2} \rangle, \langle s_{i,2} \text{ Val } a_{i,2} \rangle, \langle w_i \text{ Has } s_{i,3} \rangle, \langle s_{i,3} \text{ Val } a_{i,3} \rangle, \langle w_i \text{ Has } p_i \rangle, \langle p_i \text{ Val } y_i \rangle, \langle u_i \text{ type } y_i \rangle$, where the name of any query variable $a_{i,j}$ coincides with the one of the j -th variable occurring in the i -th clause of F' , and ϕ_2 is composed of the atoms $\langle c_{z,1} \text{ In } w_z \rangle, \langle w_z \text{ Has } s_{z,1} \rangle, \langle s_{z,1} \text{ Val } z \rangle, \langle w_z \text{ Has } s_{z,2} \rangle, \langle s_{z,2} \text{ Val } \bar{z} \rangle, \langle w_z \text{ Has } p_{z,1} \rangle, \langle p_{z,1} \text{ Val } y_{z,1} \rangle, \langle w_z \text{ Has } p_{z,2} \rangle, \langle p_{z,2} \text{ Val } y_{z,2} \rangle, \langle u_z \text{ type } y_{z,1} \rangle, \langle y_{z,2} \text{ subClassOf } E \rangle$.

Intuitively, ϕ_1 is used to check for the existence of atoms in the derived definite graph encoding an assignment to the variables of clause c_i in such a way that the assignment makes c_i true, while ϕ_2 is used to check the existence of atoms in the derived definite graph encoding an assignment ensuring that one variable between z and $\bar{z}$ is true.

D Proof of Theorem 2

$\Rightarrow$ Suppose that F' is satisfiable. We prove that $G^{F'} \models Q^{F'}$ by showing that, for every definite RKG derived from $G^{F'}$, the answer to $Q^{F'}$ is true. Consider an assignment μ for the universally quantified variables of F' , and let $K_\mu^{F'}$ be the RKG derived from $G^{F'}$ and such that, for each x_i to which μ assigns false, the axiom $\langle x_i \text{ subClassOf } E \rangle$ belongs to $G_d^{F'}$, and for each x_j to which μ assigns false, an axiom of the type $\langle e_j \text{ type } x_j \rangle$ for some e_j . Since we are assuming that F' is satisfiable, then there exists an assignment ν for the existential variables of F' such that the assignment $\mu \cup \nu$ makes F' true. Consider the following binding for the variables of the query $Q^{F'}$:

- For each clause $c_i = (z_1, z_2, z_3)$ and the associated subquery ϕ_1 of $Q^{F'}$, w_i is bound to t_{l_1, l_2, l_3}^i where l_1, l_2, l_3 correspond to the truth assignments provided by $\mu \cup \nu$ to the variables of c_i , while $s_{i, h}$ is bound to either z_h or to $v_{l_1, l_2, l_3}^{i, h}$ depending on whether z_h is a universally quantified variable (first binding) or not (second binding). In the first case, z_h is bound to itself, while in the second case it is bound to either D , if $l_h = 1$, or to E otherwise. The variables p_i and y_i are bound to the same values as $s_{i, h}$ and z_h for h such that $l_h = 1$ (there exists at least one, by construction of $G^{F'}$).
- For each variable z of F' and the associated sub-query ϕ_2 , w_p is bound to t_{j_1, j_2}^z such that l_1, l_2 correspond to the assignment given by $\mu \cup \nu$ to z and $\bar{z}$. Variable $s_{z, 1}$ is bound to z if z is universally quantified, and in this case the query variable z is bound to itself; otherwise $s_{z, h}$ is bound to $v_{l_1, l_2}^{z, h}$; if $l_h = 1$, z and $\bar{z}$ are bound to D and to E , respectively, while if $l_h = 0$ they are bound to E and to D , respectively. If $l_1 = 1$, query variables $p_{z, 1}$ and $p_{z, 2}$ are bound to z and $\bar{z}$, respectively, while they are bound to $\bar{z}$ and to z otherwise.

The binding considered above makes $Q^{F'}$ true for $K_d^{F'}$ under the given assignment.

$\Leftarrow$ Suppose that $G^{F'} \models Q^{F'}$. We show that F' is satisfiable. The premise implies that, for any class x_j corresponding to the universally quantified variable x_j of F' , there exists a binding making the query true. It is easy to verify that, starting from any such binding, it is possible to derive an assignment for existentially quantified variables such that it makes F' true. $\square$